ECS7032P Assignment 1 – Individual Reflection

Hakan Coskun · Student no. 241076864 · Group W

Group report: Decoupled Information-Set MCTS with a Card-Value Model for Sushi Go!

As group leader I organised the project and divided the work so that each of us carried an equal share. Early on we agreed to build on BasicMCTS, the permitted base agent, with every modification as a switch, so that one code base supports the ablations. With the group I set the research questions (which adaptations make the agent stronger, and how their value changes with the time per decision and the number of players) and planned the experimental protocol: 500 games per condition, 95% intervals, budgets from 100 to 900 ms, three and four players, and a final validation on deals never used in development. I also kept the report one coherent story rather than three separate projects.

Halit turned our plan into working code, ran the experiments and prepared the competition files, while Weronika's research on hidden information and simultaneous moves gave the agent its technical direction.

The most useful lesson was how much evidence a claim needs: with 500 games a win rate still has an interval of about ±4.4 percentage points, so several promising changes turned out to be noise. Planning the held-out validation paid off: our agent won 52.6% of three-player games on unseen deals, within sampling error of our earlier results. Next time I would fix the deal sets earlier, as our first comparisons used different deals and could not be paired.

Use of generative AI: as our report states, we used Claude (Anthropic) for the code and the drafting; this reflection was also drafted with Claude from our own account of the work.

Word count: 260 (without the heading)